

THE KENYA NATIONAL EXAMINATION COUNCIL

KCPE 2009

654321

MATHEMATICS

READ THESE INSTRUCTIONS CAREFULLY

Time : 2 hours

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 question.
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet.

HOW TO USE THE ANSWER SHEET

4. Use only an ordinary pencil.
5. Mark sure that you have written on the answer sheet:

YOUR INDEX NUMBER**YOUR NAME****NAME OF YOUR SCHOOL**

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number (i.e. School Code Number and the three figured Candidate's Number) in the grid near the top of the answer sheet.
7. Do not make any marks outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answer are given. Then answer are lettered A,B,C and D. In each case only ONE of the answers is correct. Choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written.

Example**In the question booklet:**

11. What is the value of $6(24-18)+6 \times 4?$

6

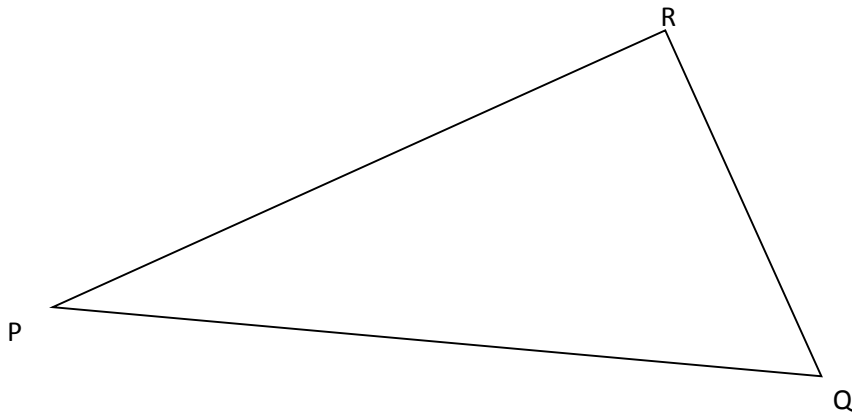
- A. 30
- B. 25
- C. 10
- D. 28

1 [A] [B] [C] [D] 11. [A] [B] [C] [D] 21 [A] [B] [C] [D] 31 [A] [B] [C] [D]

12. Your dark line **MUST** be within the box
13. For each question **ONLY ONE** box is to be marked in each set of four boxes.

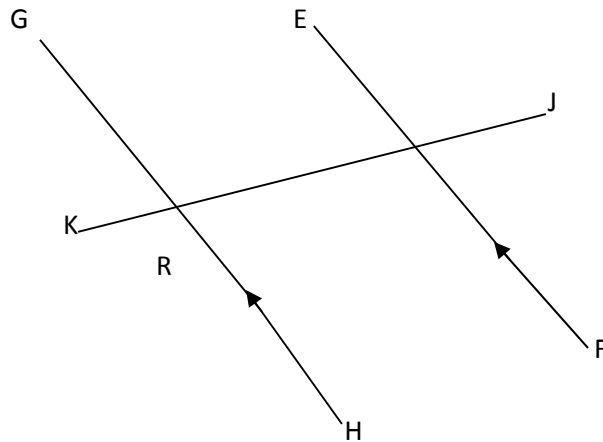
The Kenyan Examination Council, 2009.

1. Which one of the following numbers is twenty six million, four hundred and thirty thousand, two hundred and nine in symbols?
 - A. 26432009
 - B. 2643209
 - C. 26430309
 - D. 2643302009
2. Which one of the following statement is correct?
 - A. $\frac{3}{4} > 0.75$
 - B. $\frac{1}{9} < 0.1$
 - C. $\frac{4}{3} < 1.3$
 - D. $\frac{2}{5} < 0.5$
3. A storekeeper donated 5 tonnes of sugar to families in the vantage. Each family receive 2.5kg of sugar.
How many families benefited?
 - A. 20 000
 - B. 2 000
 - C. 200
 - D. 20
4. On the diagram below construct a perpendicular from R to cut PQ at S?



- A. 4.9cm
- B. 4.1cm
- C. 5.2cm
- D. 5.8cm

5. On the diagram below EF is parallel to GH and JK is a transversal. Which one of the angles is equal to KRH?



- A. SRH
B. GRK
C. ESJ
D. JSF
6. Mwende had mangoes. She ate 5 and shared the remaining among her 6 friends. She however found that she needed 2 more mangoes. How many mangoes had she at the beginning?
- A. 22
B. 27
C. 29
D. 31
7. Maloba deposited Sh. 8000 in a financial institution that paid simple interest. At the end of 3 years this money had amounted to Sh. 9200. At what rate per annum was the simple interest awarded?
- A. 5%
B. 15%
C. $38\frac{1}{2}\%$
D. $4\frac{8}{23}\%$
8. Adhiambo left home and cycled for $1\frac{1}{2}$ hours at an Average speed of 8 km/h. She rested for 30 minutes and continued with the journey for 2 hours at an average speed of $7\frac{1}{2}$ km/h. What was the average speed for the whole journey.
- A. 27km/h
B. $7\frac{5}{7}$ km/h
C. $7\frac{3}{4}$ km/h
D. $6\frac{3}{4}$ km/h
9. The bar graph shows the number and the sizes of pairs of the shoes sold in one week.

Which size of the shoe was brought most?

- A. 12
- B. 11
- C. 8
- D. 7

10. What is the difference between the total value of digit 3 and digit 9 in the number 6803901?

- A. 6
- B. 2100
- C. 3000
- D. 3900

11. What is the value of

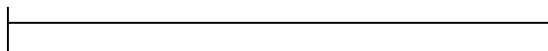
$$\frac{6(24 - 18) + 6 \times 4}{6}$$

- A. 30
- B. 25
- C. 10
- D. 28

12. Juma slept at 2130h. After sleeping for 8h 45 minutes he work up. At what time in am/pm, did he work up?

- A. 6.15am
- B. 12.15am
- C. 12.15am
- D. 6.15pm

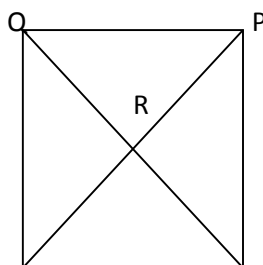
13. Using line PQ below, construct a parallelogram PQRS in which QR = 7.5cm
Angle SPQ = 60°. Join PR.



What is the size of angle PRS

- A. 120°
- B. 40°
- C. 30°
- D. 20°

14. The figure below is a square in which the diagonals intersect at R



Which of the following statement is true for triangle MRN?

- A. All sides are equal
- B. One angle is angle is 60°
- C. C. All angles are equal
- D. One angle is right angle

15. If a shopkeeper sells a basin for sh. 72, he would make a loss of 10%. At what price must he sell the basin so that he makes a profit of 20%

- A. Sh. 96
- B. Sh. 86.40
- C. Sh. 80
- D. Sh. 79.20

16. What is the value of $\frac{2w(x-2)^2}{y+1}$ when

$x = 5$, $y = x + 3$ and $w = 2x + y$?

- A. 12
- B. 24
- C. 36
- D. 144

17. Below are two graphs. One graph shows the journey by a bus from Nairobi to Nakuru. The other shows the journey by a car from Nakuru to Nairobi.

How far from Nakuru was the bus when the car reached Nairobi?

- A. 115km
- B. 90km
- C. 60km
- D. 35km

18. In a meeting $\frac{3}{5}$ of the attendants were women, $\frac{1}{4}$ of the remainder were men and the rest were children. There were 80 men in that meeting. How many children were there?

- A. 120
- B. 240
- C. 480
- D. 800

19. What is the number 15.76 rounded off to the nearest tenth?

- A. 15.7
- B. 15.8
- C. 16
- D. 20

20. What is the value of

$$\frac{4}{5} + \frac{3}{5} \text{ of } \left(\frac{2}{3} - \frac{1}{5} \right) + \frac{3}{4} \times \frac{2}{5}?$$

- A. $1\frac{19}{50}$
- B. $1\frac{143}{150}$
- C. $1\frac{3}{10}$
- D. $1\frac{23}{250}$

21. A school contributed money for project as follows: Pupils in class 1 -4 contributed sh. 10 each, Pupils in class 5 -8 contributed sh. 20 each Teachers contributed sh. 200 each The head teacher contributed sh. 500. Each class in school was double streamed with 30 pupil per stream. The number of teachers included the head teacher was 26. How much money was contributed altogether?

- A. Sh. 12 700
- B. Sh. 15 100
- C. Sh. 19 900
- D. Sh. 20 100

22. The table below shows the fare in shillings for a matatu travelling from Nairobi to Nakuru.

Fourteen passengers boarded the matatu at Nairobi. Seven of the Passengers alighted at Gilgil while five others boarded at Gilgil for Nakuru. How much money altogether did the driver collect?

- A. Sh. 3650
- B. Sh. 3500
- C. Sh.3150
- D. Sh. 1900

23. The table below shows the number of pupils who were in standards 1 to 4 in a certain school from 2001 to 2004

	Std 1	Std 2	Std 3	Std 4
2001	79	73	59	55
2002	83	77	71	57
2003	86	80	74	68
2004	89	83	77	71

How many pupils of the class which was in Std 1 in 2001 had dropped out of the class by 2004?

- A. 24
- B. 16
- C. 10
- D. 6

24. Matu paid sh. 7600 for a radio after getting a 5% discount on the marked price. How much would he have paid if he had been given a 15% discount?

- A. Sh. 8000
- B. Sh. 6800

C. Sh. 6840

D. Sh. 6460

25. What is the value of s in the equation

$$\frac{5}{4} + \frac{2(s+5)}{3} = 6?$$

A. $2\frac{10}{11}$ B. $1\frac{1}{11}$ C. $10\frac{2}{11}$ D. $2\frac{4}{9}$

26. The bar graph below the number of pupils in six classes in a school.

What is the mean number of pupil per class?

A. 26

B. 38

C. 41

D. 246

27. What is the next number in the sequence

6, 13, 24, 37, _____?

A. 58

B. 56

C. 54

D. 52

28. A rectangular container is 80cm long, 50cm wide and 40 cm high. The container is filled with water to a level 30cm high. What is the volume of empty space in the container?

A. $180\,000\text{cm}^3$ B. $160\,000\text{cm}^3$ C. $120\,000\text{cm}^3$ D. $40\,000\text{cm}^3$

29. A team of eleven players took lunch in a hotel, which offered food prices as shown in the table below.

	Ugali	Rice	Chips
Beef	Sh. 60	Sh. 65	Sh. 75
Chicken	Sh. 70	Sh. 70	Sh. 100
Fish	Sh. 75	Sh. 80	Sh. 105

5 players each took ugali with chicken

3 players each took chips with fish

The remaining players each took rice with beef. Musa gave one thousand shilling note to pay for the lunch of all the players. What balance did he get?

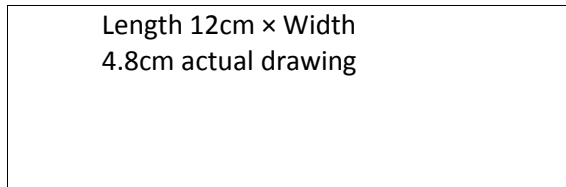
A. Sh. 55

B. Sh. 115

C. Sh. 310

D. Sh. 885

30. The diagram below is a scale drawing of a plot of land draw to the scale 1 : 500



What is the actual area of the plot, in m²?

- A. 600 000
- B. 90 000
- C. 13 500
- D. 510

31. What is the value of $\frac{2.8 - 0.5 \times 3.2 + 3}{0.4}$?

- A. 4.2
- B. 1.8
- C. 10.5
- D. 25.9

32. Sera sells googs for a company. She is paid a salary of sh. 84 000 per month plus a commission of 20% for the sale of goods worth above sh. 10 000. In one month her total earing were sh. 150 000. What was the value of the goods she sold?

- A. Sh. 760 000
- B. Sh. 340 000
- C. Sh. 330 000
- D. Sh. 244 000

33. A father shared money among his three children Mulwa, Wangare and Achesa. Wangare received sh. 10 more than Achesa, While Mulwa got twice the amount of Wangare got. If Achesa received sh. X, which one of the following expression represents the total amount of the money given to the three children?

- A. $4x + 30$
- B. $4x + 30$
- C. $2x + 20$
- D. $3x + 30$

34. In a class the ratio of the number of boys to girls is 2 : 3. What is the percentage of the number of boys in the class?

- A. $33 \frac{1}{3} \%$
- B. 40%
- C. 60%
- D. $66 \frac{2}{3}\%$

35. The diagram below is a net of a solid.

If the net is folded, which one of the following diagrams represent the solid formed?

36. The figure below represents a vegetable garden consisting of a square and four semicircles each of diameter 4.2m.



What is the area of the garden in square metres? (Take $\pi = 22/7$)

- A. 69.6
- B. 225
- C. 252.72
- D. 333.88

37. What is the difference between the L.C.M. and the G.C.D. of the number 24, 32 and 40?

- A. 8
- B. 472
- C. 480
- D. 488

38. A school offers only 4 games. Each pupil was asked to choose one favourite game out of the 4.

0.4 of the pupils chose football

0.2 of the pupils chose netball

0.3 of the pupils chose basketball

The rest, who were 112 pupils, chose hockey.

What was the total number of the pupils who chose football and netball?

- A. 240
- B. 160
- C. 400
- D. 80

39. A sales man had 2 240 litres of juice which she packed in 500ml and 250ml packets. Three quarters of the juice was packed in 500ml packets and the rest in 250ml packets.

What was the total number of the packets used?

- A. 56

- B. 560
- C. 5600
- D. 56000

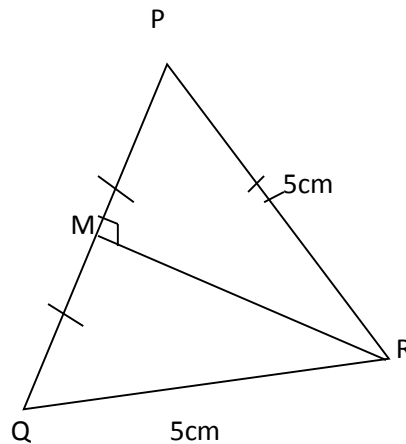
40. A rectangular plot of the land has a perimeter of 35m. The longer side is 10m. A wall is to be constructed along one of the diagonals of the plot. What would be the length of the wall?

- A. $78 \frac{1}{8}m$
- B. $171 \frac{1}{2}m$
- C. $7 \frac{1}{2}m$
- D. $12 \frac{1}{2}m$

41. Mary bought a T.V. a set on hire purchase terms. She paid a deposit of sh. 800 and 15 equal monthly instalments of sh. 700 each. The hire purchase price was 25% higher than the cash price. Amina bought the same T.V. set on cash. How much more than Amina did Mary pay for the T.V. set?

- A. Sh. 2 260
- B. Sh. 2 825
- C. Sh. 9 040
- D. Sh. 11 300

42. In triangle PQR below, $PQ = 8cm$, $PM = MQ$ and $PR = RQ = 5cm$. What is the area of the triangle PQR?



- A. $7 \frac{1}{2} cm^2$
- B. $6cm^2$
- C. $12 \frac{1}{2} cm^2$
- D. $12cm^2$

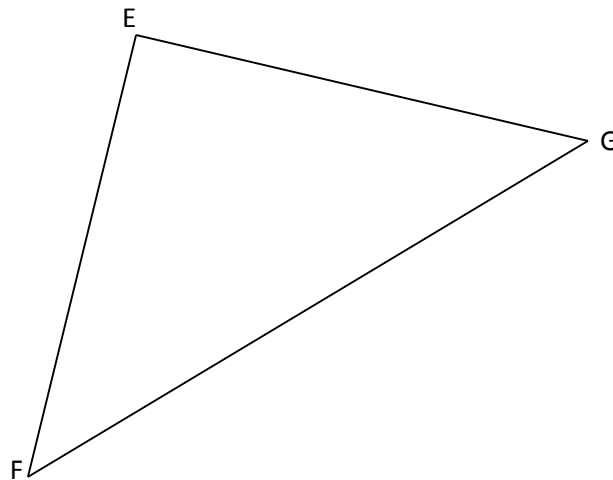
43. Eighteen people can take 72 days to complete the piece of work. If 2 of the people did not turn up for the work, how many more days would the remaining number of people, working at the same rate take to complete the work?

- A. 9
- B. 81
- C. 153
- D. 575

44. Pupils in classes 5,6,7 and 8 to raise funds for a party. Class 7 raised sh. X while class 5 raised $\frac{1}{3}$ of what class 7 raised. Class 6 raised sh. 100 less than the total amount raised by both classes 5 and 7. If the total amount was sh. 6 900, which one of the following equation can be used to find the amount raised by each class?

- A. $X + \frac{2}{3}x + 100 = 6900$
- B. $3x + \frac{2}{3}x + 300 = 6900$
- C. $3x \frac{2}{3}x + 300 = 6900$
- D. $3x + \frac{2}{3}x + 100 = 6900$

45. On the triangle EFG below construct the bisector of angle EFG to meet the line EG at M. construct the bisector of the Angle FEG to meet the line FG at N. Mark P at the point where the two bisectors intersect.



What is the size of the angle EPF?

- A. 113°
- B. 67°
- C. 48°
- D. 20°

46. What is the value of

$$\sqrt{\frac{1}{4} \div \frac{2}{9} \times \left(\frac{2}{3}\right)^2}$$

- A. $\frac{6}{25}$
- B. $1 \frac{1}{9}$
- C. $\frac{11}{5}$
- D. $\frac{2}{5}$

47. In year 2005, enrolment in a school was 450. In the year 2006, the enrolment decreased by 10% from that of year 2005. The enrolment increase by 20% in year 2007. What was the enrolment in year 2007?
- A. 396
 - B. 486
 - C. 495
 - D. 584
48. A cylinder tank of the diameter 105 cm. What is the height of the water in the tank? (Take π 22/7)
- A. 5.6cm
 - B. 56cm
 - C. 224cm
 - D. 1470cm
49. How many blocks can be used to make the pile shown in the diagram below?
- A. 63
 - B. 60
 - C. 54
 - D. 36